

GAS_LEAK_EMERGENCY_SOP

STEEL AUTHORITY OF INDIA LIMITED
BOKARO STEEL PLANT

Standard Operating Procedure

SOP No: BSP-GS-004

Title: Emergency Response Procedure for Gas Leak

Department: Safety and Environment

Revision: 03

Date: 20-Mar-2024

1. PURPOSE

To define the immediate response actions for all personnel upon detection of a gas leak at Bokaro Steel Plant, covering Blast Furnace Gas (BFG), Coke Oven Gas (COG), and Linz-Donawitz Gas (LDG).

2. SCOPE

Applicable to all areas of Bokaro Steel Plant where process gases are present. Mandatory for all plant personnel, contractors, and visitors.

3. GAS HAZARD CLASSIFICATION

- Blast Furnace Gas (BFG): CO content 20-28%. Highly toxic. Odourless.
- Coke Oven Gas (COG): Contains H₂, CH₄, CO. Flammable. Toxic.
- LD Gas (LDG): CO content 60-70%. Extremely toxic. Odourless.

CRITICAL: BFG and LDG are odourless. Never rely on smell for detection. Always use gas detectors.

4. DETECTION THRESHOLDS

- CO alarm (warning): 20 ppm
- CO alarm (evacuate): 50 ppm
- Explosive limit for COG: 6% in air (Lower Explosive Limit)

5. IMMEDIATE RESPONSE PROCEDURE

Step 1 — On Detection of Gas Leak

- Do NOT panic
- Do NOT operate any electrical switches (risk of ignition)
- Do NOT use mobile phones in the affected area
- Raise alarm immediately using nearest manual call point
- Call Emergency Control Room: Extension 100

Step 2 — Evacuation

- Evacuate all non-essential personnel from the affected zone immediately
- Move upwind of the leak source
- Minimum safe distance: 100 metres from leak point

- Do NOT re-enter the area under any circumstances without clearance
- Assist any injured or unconscious personnel only if you have SCBA

Step 3 — Isolate the Source (Trained Personnel Only)

- Only Gas Safety Officers and authorized maintenance crew may attempt isolation
- Don SCBA before approaching the leak zone
- Identify and close the nearest upstream isolation valve
- If valve is inaccessible, contact control room for remote isolation

Step 4 — Notify

- Emergency Control Room: Extension 100 (immediate)
- Gas Safety Department: Extension 2310
- Shift Manager: Extension 2201
- Medical Centre: Extension 2400
- Fire Station: Extension 101

Step 5 — First Aid for Gas Exposure

- Remove affected person to fresh air immediately
- If unconscious, place in recovery position
- Administer oxygen if available and trained to do so
- Do NOT perform mouth-to-mouth resuscitation for CO poisoning (risk of secondary poisoning to rescuer)
- Call Medical Centre immediately: Extension 2400

6. RE-ENTRY PROTOCOL

- Re-entry permitted only after written clearance from Gas Safety Officer
- CO level must be below 20 ppm at all entry points (verified by instrument)
- Minimum two-person rule applies for any re-entry
- SCBA mandatory until continuous monitoring confirms safe levels

7. REPORTING

- All gas incidents must be reported within 1 hour to Safety Department
- Complete Incident Report Form ISF-07 within 24 hours
- Near-miss incidents must also be reported — no punitive action for reporting

8. TRAINING REQUIREMENT

All plant personnel must complete Gas Safety Training (GST-01) annually.
New joiners must complete GST-01 within 30 days of joining.